

PageRank Report

Amir Reza Khatibi
s360765

Contents

A	Introduction	2
B	Exercise 1: Adding page 5	4
C	Exercise 2:	5
D	Exercise 3:	6
E	Exercise 4:	7
F	Exercise 5:	8
G	Exercise 7:	8
H	Exercise 8:	9
I	Exercise 9:	10
J	Exercise 11:	10
	J.1 Observing the results for different values of m:	11
K	Exercise 12:	11
L	Exercise 13:	12
M	Exercise 14:	13
N	Exercise 16:	14
O	Exercise 17:	15

A Introduction

PageRank is a well-known algorithm developed by Google to rank web pages based on their relative importance. The algorithm models the web as a directed graph where each page is represented as a node and links between pages are represented as edges. The PageRank score of a page is based on the number and quality of backlinks it has:

1. We first set

$$x_k = \# \text{ of backlinks of page } k$$

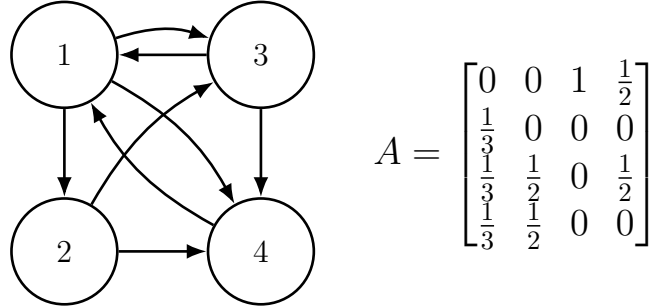
2. Then we update

$$x_k = \sum_{j \in L_k} \frac{x_j}{n_j}$$

where n_j is the number of outgoing links from page j .

We then try to form an $n \times n$ **column stochastic matrix** A (the sum of the entries in each column is equal to one), where the element A_{ij} represents the probability of moving from page j to page i .

Example of A for a graph:



To find the importance (rank) of each page, we seek the **dominant eigenvector** corresponding to eigenvalue 1 of the matrix A (every column-stochastic matrix has 1 as an eigenvalue).

The **power method** is an iterative algorithm used to compute the *dominant eigenvector* (the one with the largest magnitude) of a matrix. Starting with an initial vector, the algorithm repeatedly multiplies the vector by the matrix and normalizes it until it converges. The power method is well-suited for large sparse matrices like the ones used in web ranking.

$$x_k = Mx_{k-1} \quad \text{therefore} \quad x_k = M^k x_0, \quad k \rightarrow \infty$$

At each step we should also normalize the result so that it does not decay to zero or grow without bound:

$$x_k = \frac{Mx_{k-1}}{\|Mx_{k-1}\|}$$

A problem arises when there is no unique eigenvector for $\lambda = 1$:

- One such case is when a web W has n pages and r component sub-webs $W_1, W_2, \dots, W_r$, which forces $\dim(V_1(A)) \geq r$.
- Another problem is the case of *dangling nodes* that have no outgoing links. In this case, A contains one or more columns of all zeros and is called *column-substochastic*.

When the web is not connected and $\dim(V_1(A)) > 1$, if there is no dangling node, one remedy is to replace A with the matrix

$$M = (1 - m)A + mS,$$

where $m \in [0, 1]$ and S is a matrix with the same size as A whose entries are all equal to $\frac{1}{n}$. Using M instead of A assigns a web page with no backlinks an importance score of $\frac{m}{n}$.

An important point is that for M **positive** and **column-stochastic**, any eigenvector in $V_1(M)$ has all positive or all negative components (proof by contradiction). Moreover, $\dim(V_1(M)) = 1$ as well (again, proof by contradiction). Therefore, our problem is solved.

Summary: The **positive column-stochastic** matrix M has a unique vector q with positive components such that

$$Mq = q,$$

where $\|q\|_1 = 1$. Moreover,

$$q = \lim_{k \rightarrow \infty} M^k x_0$$

for any x_0 with positive components such that $\|x_0\|_1 = 1$. The vector q may also be computed as the limit of the iterations

$$q = \lim_{k \rightarrow \infty} x_k = (1 - m)Ax_{k-1} + ms,$$

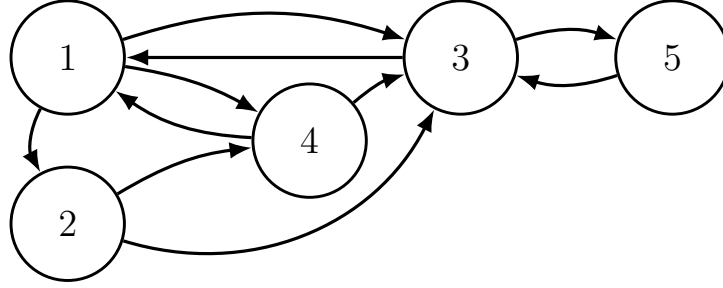
which is a far more efficient computation since M has no zero elements, while A can be expected to contain mostly zeros (most web pages link to only a few other pages).

The aim of this report is to study the PageRank algorithm from both a theoretical and a computational perspective. In particular, we analyze how the structure of the web graph influences the ranking of pages, and we highlight the limitations of the basic PageRank formulation based solely on the link matrix A , such as the presence of dangling nodes, reducible graphs, and non-uniqueness of the dominant eigenvector. To address these issues, we introduce the modified matrix $M = (1-m)A + mS$ and investigate the role of the damping parameter m in ensuring positivity, uniqueness, and stability of the ranking. Finally, we study the convergence behavior of the power method and relate the observed numerical results to spectral properties of M , including the second largest eigenvalue and the Perron–Frobenius theorem.

B Exercise 1: Adding page 5

To find the importance (rank) of each page, we seek the **dominant eigenvector** corresponding to eigenvalue $\lambda = 1$ of the matrix A

The graph obtained after adding page 5 will be:



Therefore the corresponding matrix will be:

$$A = \begin{bmatrix} 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{3} & 0 & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{2} & 0 & \frac{1}{2} & 1 \\ \frac{1}{3} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$$

The computed e is then:

$$e = [0.244, 0.081, 0.367, 0.122, 0.183],$$

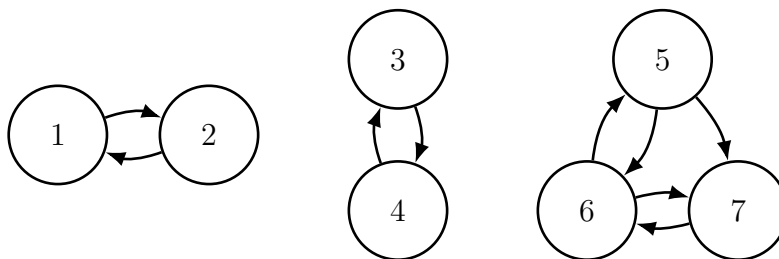
indicating that the importance score of page 3 is now greater than that of page 1. It is reasonable since page 3 does not link only to page 1 anymore so it does not cast its entire vote for page 1

C Exercise 2:

In this exercise, we would like to consider a web graph with three subwebs showing that

$$\dim(V_1(A)) \geq 3,$$

which equals (or exceeds) the number of connected components in the web. Here, $V_1(A)$ denotes the eigenspace of A corresponding to the eigenvalue $\lambda = 1$.



The computed list of eigenvalues is then:

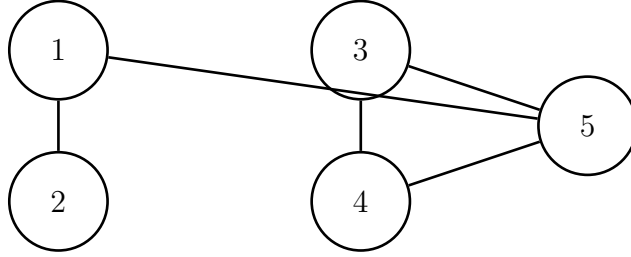
$$\textit{lambda}s = [1, -1, 1, -1, 1, -0.5, -0.5],$$

which shows that

$$\dim(V_1(A)) = 3,$$

D Exercise 3:

In this exercise we add a link from page 5 to page 1 in the second figure of the table and consider the graph as an undirected graph which makes it connected. The obtained graph is:



The corresponding matrix is:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & \frac{1}{3} \\ \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{3} \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

and the corresponding list of eigenvalues is:

$$\text{lambda}s = [-0.86, -0.5, -0.30, 0.65, 1],$$

which makes

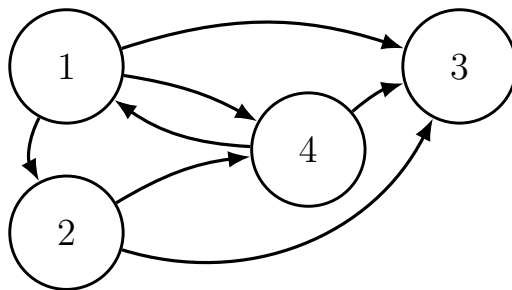
$$\dim(V_1(A)) = 1,$$

This result makes sense since after connecting nodes 5 and 1, the graph is connected and we don't have two sub-webs, so

$$\dim(V_1(A)) \geq 1$$

E Exercise 4:

In this exercise, we consider a graph with a dangling node, we would like to find its largest positive (Perron) eigenvalue and a non-negative Perron eigenvector scaling it so that components sum to one. The graph is:



which gives us the **column-substochastic matrix** A:

$$A = \begin{bmatrix} 0 & 0 & 0 & \frac{1}{2} \\ \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{2} & 0 & 0 \end{bmatrix}$$

The resulting Perron eigenvalue is $\lambda = 0.56$ and the normalized Perron eigenvector is

$$x = [0.21, 0.12, 0.44, 0.23],$$

which shows that page 3 is the most important page. It makes sense due to the fact that it is pointed by all other pages and doesn't point to any other page, so it doesn't distribute its mass at all.

Also note that since A is substochastic the top eigenvalue drops below 1!

F Exercise 5:

In this exercise we would like to prove that in any web the importance score of a page with no backlinks is zero [$L_k = \emptyset$, where k is a page with no backlinks]

The importance score is defined as:

$$x_k = \sum_{j \in L_k} \frac{x_j}{n_j}.$$

In the link matrix A , we have $A_{kj} > 0$ if page j links to page k . If page k has no backlinks, then row k of A is identically zero. Thus, since $x = Ax$, it follows that

$$x_k = (Ax)_k = \sum_{j=1}^n A_{kj}x_j = 0.$$

G Exercise 7:

In this exercise, we would like to prove that if A is an $n \times n$ column-stochastic matrix and $0 \leq m \leq 1$, then

$$M = (1 - m)A + mS$$

is also a column-stochastic matrix.

A matrix is column-stochastic if all of its entries are non-negative and the entries in each column sum to one:

1. **Non-negativity.** For all i, j we have

$$M_{ij} = (1 - m)A_{ij} + mS_{ij} \geq 0,$$

since $A_{ij} \geq 0$, $S_{ij} \geq 0$, and $0 \leq m \leq 1$.

2. **Column sums.** We show that each column of M sums to one. Fix any column j . Then

$$\sum_{i=1}^n M_{ij} = \sum_{i=1}^n ((1 - m)A_{ij} + mS_{ij}) = (1 - m) \sum_{i=1}^n A_{ij} + m \sum_{i=1}^n S_{ij}.$$

Since A and S are column-stochastic matrices, we have

$$\sum_{i=1}^n A_{ij} = 1 \quad \text{and} \quad \sum_{i=1}^n S_{ij} = 1.$$

Therefore,

$$\sum_{i=1}^n M_{ij} = (1 - m) + m = 1.$$

H Exercise 8:

In this exercise, we would like to Show that the product of two column-stochastic matrices is also column-stochastic.

Let A and B be two $n \times n$ column-stochastic matrices, that is,

$$A_{ij} \geq 0 \quad \text{and} \quad B_{ij} \geq 0 \quad \text{for all } i, j,$$

and for each column j ,

$$\sum_{i=1}^n A_{ij} = 1 \quad \text{and} \quad \sum_{i=1}^n B_{ij} = 1.$$

Let $C = AB$.

1. **Non-negativity.** For all i, j ,

$$C_{ij} = \sum_{k=1}^n A_{ik} B_{kj},$$

which is a sum of non-negative terms. Hence,

$$C_{ij} \geq 0.$$

2. **Column sums.** Fix any column j . Then

$$\sum_{i=1}^n C_{ij} = \sum_{i=1}^n \sum_{k=1}^n A_{ik} B_{kj} = \sum_{k=1}^n \left(\sum_{i=1}^n A_{ik} \right) B_{kj}.$$

Since A is column-stochastic, we have

$$\sum_{i=1}^n A_{ik} = 1 \quad \text{for all } k,$$

and therefore

$$\sum_{i=1}^n C_{ij} = \sum_{k=1}^n B_{kj} = 1,$$

because B is column-stochastic.

Since C is non-negative and each of its columns sums to one, $C = AB$ is column-stochastic.

I Exercise 9:

In this exercise we would like to demonstrate that a page with no backlinks is given importance score $x = \frac{m}{n}$ using the formula $x = (1 - m)Ax + ms$

If page k has no backlinks, then in the link matrix A the entire row k is zero, that is,

$$A_{kj} = 0 \quad \text{for all } j.$$

Therefore,

$$(Ax)_k = \sum_{j=1}^n A_{kj}x_j = 0.$$

Consequently, using the PageRank equation

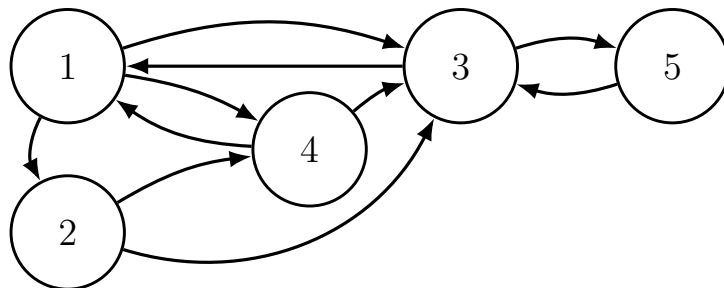
$$x = (1 - m)Ax + ms,$$

we obtain

$$x_k = (1 - m)(Ax)_k + ms_k = \frac{m}{n}.$$

J Exercise 11:

In this exercise we would like to compute the ranking of the pages for the graph of exercise 1:



But this time we want to use the matrix

$$M = (1 - m)A + mS$$

with $m = 0.15$

The eigenvector of M corresponding to $\lambda = 1$ is then calculated as:

$$e = [0.24, 0.09, 0.35, 0.14, 0.18]$$

which gives the ranking for the pages: $3 > 1 > 5 > 4 > 2$

Using matrix M , at each step, our "random surfer" does:

- with probability $1-m$: follow a hyperlink according to A
- with probability m : teleport to a uniformly random page (That's S)

J.1 Observing the results for different values of m :

for $m = 0$,

$$e = [0.24, 0.09, 0.37, 0.12, 0.18]$$

for $m = 0.3$,

$$e = 0.23, 0.11, 0.33, 0.15, 0.18$$

for $m = 0.6$,

$$e = [0.21, 0.15, 0.28, 0.177, 0.176]$$

for $m = 1$,

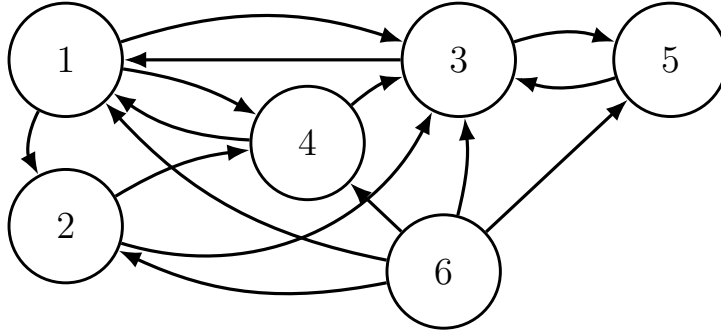
$$e = [0.2, 0.2, 0.2, 0.2, 0.2]$$

It can then be observed that increasing m increases the "teleportation" effect which smooths the PageRank distribution by pushing toward $\frac{1}{n}$; thus high-ranked pages lose some weight while low-ranked pages gain weight, and for $m = 1$ the ranking becomes uniform and independent of the graph.

Also, the rankings might change by changing the value of m , you can see that with $m = 0.6$, page 4 actually surpasses page 5 and becomes more important!

K Exercise 12:

In this exercise we would like to compare the results obtained using matrix A and using matrix M with $m = 0.15$ for the graph below:



Comparison between using A and using M

$$e_A = [0.245, 0.081, 0.367, 0.122, 0.184, 0],$$

$$e_M = [0.231, 0.095, 0.340, 0.135, 0.174, 0.025].$$

Using A , page 6 receives score 0 because it has no backlinks (its incoming-link set is empty), hence it receives no contribution from the update $x_k = \sum_{j \in L_k} x_j / n_j$. In contrast, using M assigns page 6 a strictly positive score ($\frac{m}{n} = 0.025$)

due to the teleportation term mS , which injects a uniform baseline mass into every page. Numerically, the damping also *smooths* the distribution: the most important pages (notably page 3 and page 1) decrease slightly, while weaker pages (such as page 2 and page 4) increase. This smoothing can also change the ordering of mid-ranked pages sometimes.

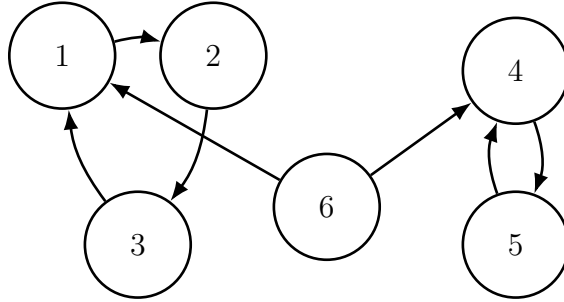
Connection to Perron–Frobenius. The matrix A is column-stochastic, so $\lambda = 1$ is an eigenvalue. However, A may be reducible and may contain zero entries; in such cases, Perron–Frobenius does *not* guarantee that the eigenvector for $\lambda = 1$ is unique and strictly positive, and some components of the eigenvector may be zero (as observed for page 6). On the other hand, for $0 < m \leq 1$ and $S = \frac{1}{n}\mathbf{1}\mathbf{1}^T$, we have

$$M_{ij} = (1 - m)A_{ij} + mS_{ij} \geq m \cdot \frac{1}{n} > 0,$$

so M is a **positive** (entry-wise) and **column-stochastic** matrix. By the Perron–Frobenius theorem, a positive matrix has a unique (up to scaling) dominant eigenvector with strictly positive entries, associated with the spectral radius. Since M is column-stochastic, its spectral radius is 1, hence $\lambda = 1$ is the dominant eigenvalue and the corresponding eigenvector is unique (after normalization, e.g. $\|e_M\|_1 = 1$) and satisfies $e_M > 0$ component-wise. This explains why introducing $m > 0$ removes zero scores and yields a well-defined, robust ranking for all pages.

L Exercise 13:

In this exercise we would like to use M to compute the rankings of a web with three sub-webs, the graph is shown below:



The obtained eigenvector is:

$$q = [0.140, 0.188, 0.243, 0.205, 0.199, 0.025]$$

This shows that different from the case of using A , here $\dim(V_1(M)) = 1$ and it is not more than or equal to the number of sub-webs.

M Exercise 14:

In this exercise we would like to compute the values of

$$\|M^k x_0 - q\|_1 \quad \text{and} \quad \frac{\|M^k x_0 - q\|_1}{\|M^{k-1} x_0 - q\|_1}$$

for $k = 1, 5, 10, 50$, using an initial guess x_0 not too close to the actual eigenvector q and determine

$$c = \max_{1 \leq j \leq n} \left| 1 - 2 \min_{1 \leq i \leq n} M_{ij} \right|$$

and the absolute value of the second largest eigenvalue of M for the web used in exercise 11. The computed q as before is:

$$q = [0.24, 0.09, 0.35, 0.14, 0.18]$$

Results for different values of k :

k	$\ M^k x_0 - q\ _1$	$\frac{\ M^k x_0 - q\ _1}{\ M^{k-1} x_0 - q\ _1}$
1	7.820×10^{-1}	5.125×10^{-1}
5	6.874×10^{-3}	3.227×10^{-1}
10	3.044×10^{-4}	5.245×10^{-1}
50	7.438×10^{-13}	6.113×10^{-1}

Table 1: Convergence results for the power iteration applied to M .

Moreover, $c = 0.94$ and $\lambda_2 = 0.611$.

Observations: From the numerical results reported in Table 1, we observe that the sequence $M^k x_0$ converges rapidly to the dominant eigenvector q of M . The quantity $\|M^k x_0 - q\|_1$ decreases monotonically as k increases, reaching values close to machine precision for $k = 50$. Moreover, the ratios $\|M^k x_0 - q\|_1 / \|M^{k-1} x_0 - q\|_1$ stabilize around a constant value, indicating linear convergence of the power method. This behavior is consistent with the theoretical prediction that the asymptotic convergence rate is governed by the second largest eigenvalue (in modulus) of M , whose computed value is $\lambda_2 \approx 0.611$.

Furthermore, it is known that for matrices of this form the second largest eigenvalue satisfies the bound $|\lambda_2(M)| \leq 1 - m$. In our case, with $m = 0.15$, this yields $|\lambda_2| \leq 0.85$, which is consistent with the computed value $\lambda_2 \approx 0.611$. While the constant $c = 0.94$ provides a valid upper bound ensuring convergence, the numerical results confirm that the actual convergence rate is dictated by $|\lambda_2|$, highlighting the stabilizing and accelerating effect of the damping term.

N Exercise 16:

In this exercise we would like to prove that $M = (1 - m)A + mS$ is not diagonalizable for $0 \leq m < 1$ having

$$A = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} \\ 1 & \frac{1}{2} & 0 \end{bmatrix}$$

Step 1: Explicit form of M . Using the definition of S , we compute M entrywise as

$$M = \begin{pmatrix} \frac{m}{3} & \frac{1}{2} - \frac{m}{6} & \frac{1}{2} - \frac{m}{6} \\ \frac{m}{3} & \frac{m}{3} & \frac{1}{2} - \frac{m}{6} \\ 1 - \frac{2m}{3} & \frac{1}{2} - \frac{m}{6} & \frac{m}{3} \end{pmatrix}.$$

Step 2: Eigenvalues of M . A direct computation of the characteristic polynomial yields

$$\det(\lambda I - M) = (\lambda - 1) \left(\lambda - \frac{m-1}{2} \right)^2.$$

Therefore, the eigenvalues of M are

$$\lambda_1 = 1, \quad \lambda_2 = \lambda_3 = \frac{m-1}{2}.$$

For $0 \leq m < 1$, the eigenvalue $\lambda_2 = (m-1)/2$ is distinct from 1 and has algebraic multiplicity 2.

Step 3: Geometric multiplicity. To determine the geometric multiplicity of λ_2 , we consider the nullspace of $M - \lambda_2 I$. Solving

$$(M - \lambda_2 I)v = 0$$

yields

$$\ker(M - \lambda_2 I) = \text{span} \left\{ \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \right\}.$$

Hence,

$$\dim \ker(M - \lambda_2 I) = 1.$$

Conclusion. The eigenvalue $\lambda_2 = (m-1)/2$ has algebraic multiplicity 2 but geometric multiplicity 1. Since the geometric and algebraic multiplicities do not coincide, the matrix M is *not diagonalizable* for all $0 \leq m < 1$.

Remark. For $m = 1$, we have $M = S$, whose eigenvalues are 1, 0, 0 with a two-dimensional eigenspace associated with 0. In this case, M is diagonalizable.

O Exercise 17:

In this exercise we would like to discuss how to select the value of m and its impact on the ranking and computation time.

How to choose m . The parameter m represents the probability of *teleportation* (jumping to a uniformly chosen page) versus following a hyperlink (with probability $1 - m$). Thus, m should be chosen as a compromise:

- If m is **too small** (close to 0), the ranking is driven almost entirely by the link structure. This can lead to non-uniqueness and instability when the web graph is reducible, nearly disconnected, or contains dangling nodes.
- If m is **too large** (close to 1), the ranking becomes almost uniform, since $M \approx S$ and the PageRank vector approaches $\frac{1}{n}\mathbf{1}$, losing meaningful dependence on the hyperlink structure.

In practice one typically chooses a moderate value (often around $m = 0.10$ – 0.20) to preserve link-based importance while ensuring robustness and uniqueness.

Effect on the rankings. As m increases, the PageRank vector becomes *more uniform*: high-ranked pages lose weight and low-ranked pages gain weight, because the term mS injects a uniform baseline into every component. In particular, pages with very few or no backlinks receive a positive score when $m > 0$, whereas for $m = 0$ they may receive score 0. Therefore, increasing m generally *reduces disparities* between pages and can change the ordering among mid-ranked pages.

Effect on computation time (convergence). The power method convergence is governed by the magnitude of the second largest eigenvalue of M :

$$\|x^{(k)} - q\| \approx C |\lambda_2(M)|^k.$$

For matrices of the form $M = (1 - m)A + mS$ (with $S_{ij} = 1/n$) one has the bound

$$|\lambda_2(M)| \leq 1 - m,$$

so increasing m increases the spectral gap $1 - |\lambda_2(M)|$ and therefore *accelerates convergence*. Hence:

- larger $m \Rightarrow$ faster convergence and fewer iterations,
- smaller $m \Rightarrow$ slower convergence and potentially more sensitivity to graph pathologies (reducibility, dangling nodes).

Conclusion. Choosing m is a trade-off between *ranking fidelity* and *numerical robustness/speed*. A moderate value of m yields a unique, strictly positive ranking with reasonable dependence on the link structure, while also guaranteeing faster and more stable computation via an improved spectral gap.